

Wentao Gao

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EDUCATION

University of Washington

Master of Science in Computer Science & Software Engineering

Graduate Certificate in Software Design & Development

Bothell WA

December 2025

June 2023

Related Coursework: Data Structures & Object-Oriented Programming, Parallel Computing in Grid and Cloud, High-Performance Computing, Mobile Computing, Systems Programming, Software Testing & Quality, Software Architecture, Software Management, Machine Learning, Signal Processing, Software Development Processes, Software Modeling Techniques

University of Washington

Bachelor of Arts

Seattle WA

June 2018

SKILLS

Programming Languages and Tools: C, C++, Python, C#, Java, JavaScript, HTML, CSS, React, Swift, Unity, Data-Oriented Technology Stack, Git, OpenGL, GLSL, HLSL, CUDA, OpenMP, MPI, MapReduce, Spark

Development Concepts: Data structures and algorithms, Object-oriented design principles, Agile/Scrum methodologies, Test Driven Development, CI/CD, Software Development Life Cycle

PROJECTS

VR Input Control System for Amputee Patients using Myoelectric Armband

Machine Learning, Signal Processing, Game Development, Unity, C#, Java, Python

- Designed and implemented a surface EMG(sEMG) armband-based VR input control system aimed at making VR based mirror therapy more accessible for amputee patients with phantom limb pain
- Used signal processing and machine learning techniques for myoelectric signals interpretation
- Utilized Inertial Measurement Unit(IMU) data and 3D mathematics for real-time limb orientation tracking and virtual limb reconstruction in VR

MyoPanel - Surface Myoelectric Signal Collection Tool

Python, PySide6, Bluetooth Low Energy, PyQtGraph, MediaPipe, OpenCV

- Developed an open-source desktop application used by a University of Washington Bothell lab team to collect sEMG and IMU datasets for machine-learning experiments, including the training pipeline for the VR Input Control System.
- Built an asynchronous acquisition and visualization pipeline that aligns 8-channel sEMG data sampled at 200 Hz with IMU data sampled at 50 Hz for consistent, configurable recording.
- Implemented labeling, cleaning, preprocessing, and export workflows for raw or processed EMG, IMU, and vision-based hand-landmark data to prepare datasets for model training and evaluation.

Phylogenetic Tree Construction with Distributed Computing

Distributed parallel computing, Java, Spark, MASS

- Implemented Java Spark versions of pairwise DNA-distance computation and UPGMA clustering in a three-person framework comparison; contributed to the team's MASS implementation
- Recorded DNA-distance runtime of 35.070 seconds versus an 86.946-second sequential baseline in the project report, about 2.48x faster for that stage in the selected benchmark run

PROFESSIONAL EXPERIENCE

Zoneberation LLC, ByteDance

Assistant IT Support

March 2025 - Now

- Inspect and validate conference-room IT equipment, diagnose failures, and apply fixes to restore equipment functionality. Document unresolved issues and troubleshooting steps in IT tickets for escalation to internal IT teams.